

Refractory: A Refractive Geometric Framework for Tumor Subtype Trajectory Analysis

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Abstract

Tumor subtype transitions encode critical information about disease progression and therapeutic resistance, yet the trajectories connecting distinct subtypes across the epigenome and transcriptome remain poorly understood. To the best of our knowledge, subtype trajectory analysis within a specific tumor type has not been treated as a computational problem. Inspired by RNA velocity and branching trajectory inference methods, we ask a higher-order question for the specific scenario of interest: what is the joint geometry of epigenomic and transcriptomic transitions between tumor subtypes, and how do these two molecular layers interact to shape the resistance landscape of subtype switching? We present Refractory, a novel computational framework that models these trajectories by jointly integrating chromatin accessibility and mRNA abundance within a Riemannian geometric framework. We propose a refractive pullback metric—inspired by Snell’s law, whereby light bends and slows at the boundary between media of differing densities—that encodes the local resistance of transitioning between subtype states. The inferred trajectories reveal paths of least joint epigenomic and transcriptomic effort, naturally capturing tumor plasticity—going beyond existing single-modality methods and assuming a flat isotropic latent space. Refractory provides a new geometric lens on tumor plasticity, with direct implications for understanding disease progression and drug resistance.

Keywords: Tumor Subtypes, Multiomics, Deep Generative Models, Representation Learning, Riemannian Geometry, Trajectory Analysis, Pullback Metric

1 Introduction

Tumors are complex biological systems driven by heterogeneous alterations across multiple molecular layers, where identifying subtype-specific patterns holds direct clinical relevance for improving treatment.

Tumor subtype transitions encode critical information about disease progression, plasticity, and therapeutic resistance, yet the trajectories connecting distinct subtypes across the transcriptome [1, 2], and epigenome remain poorly understood—a recent work shows the importance of understanding the epigenetic regulation across different cancers [3]. However, to the best of our knowledge, subtype trajectory analysis within a specific disease or cancer type has not been treated as a computational problem. In this work, we aim to solve this problem. We also note that the integration of multiomic data—spanning transcriptomic and epigenomic layers—has proven particularly powerful in tumor analysis [4–10], enabling a more comprehensive view of the molecular mechanisms underlying subtype diversity than any single modality alone. Motivated by this, we aim to design a trajectory inference framework that equips the multiomics shared latent space with a geometry, where distances reflect biological transition resistance rather than mere statistical dissimilarity.

Existing approaches such as RNA velocity [11]—which infers the rate of transcriptional change from the ratio of unspliced to spliced mRNA—and branching trajectory inference methods [12] reveal the direction and topology of cell fate decisions within a single modality. We ask a higher-order question: what is the joint geometry of epigenomic and transcriptomic transitions between phenotypic states, and how do these two molecular layers interact and constrain each other to shape the resistance landscape of subtype switching? Answering this requires moving beyond diffusion-based or graph-based trajectory models that assume a flat, isotropic latent space, toward a Riemannian framework with a *metric tensor*: a position-dependent matrix $g(z)$ that stretches and compresses distances differently at each point, encoding the local cost of joint regulatory-expression change. Under this geometry, *geodesics*—the generalisation of straight lines to curved spaces, defined as curves of minimal length under the metric—are not merely the shortest paths between two phenotypic states, but the paths of least biological resistance, bending around regulatory barriers and slowing near subtype boundaries where the epigenomic-transcriptomic landscape resists transition most strongly. Pulling back a metric to the latent space is a recent cutting-edge direction—we refer to it as *geometric representation learning*—that enriches the parameter space rather than relying purely on data-driven modeling, captures geometric structure, mitigates unidentifiability, and enhances uncertainty estimation beyond standard generative models [13–17].

Building on the growing field of geometric representation learning, we present Refrajectory, a novel computational framework that models tumor subtype trajectories by jointly integrating epigenomic accessibility and transcriptomic expression within a unified Riemannian geometric framework. Rather than treating gene expression space as flat or analyzing the epigenome and transcriptome in isolation, we propose a *refractive pullback metric* that aggregates chromatin accessibility signals and mRNA abundance to encode the resistance of transitioning between subtype states—analogueous to how light refracts when traversing media of different densities.

Geodesic trajectories under this metric reveal paths of least epigenomic and transcriptomic effort connecting tumor subtypes, naturally capturing branching structures and transient intermediate states that arise during tumor evolution.

2 Methods

2.1 Riemannian Geometry

As we apply Riemannian geometry to autoencoder networks, we adapt all definitions to this context. Let the encoder of the autoencoder map each observation $\mathbf{x} \in \mathbb{R}^D$ to a point $\mathbf{z} \in \mathcal{Z} \subset \mathbb{R}^d$ in a low-dimensional latent space, where $d \ll D$. We treat $\mathcal{Z}$ as a smooth Riemannian manifold $(\mathcal{Z}, g)$, equipped with a metric tensor $g(\mathbf{z})$: a symmetric positive definite matrix $g(\mathbf{z}) \in \mathbb{R}^{d \times d}$ that varies smoothly with position $\mathbf{z}$ and defines a local inner product on the tangent space $T_{\mathbf{z}}\mathcal{Z}$. For two tangent vectors $\mathbf{u}, \mathbf{v} \in T_{\mathbf{z}}\mathcal{Z}$, the inner product is:

$$\langle \mathbf{u}, \mathbf{v} \rangle_{g(\mathbf{z})} = \mathbf{u}^\top g(\mathbf{z}) \mathbf{v}. \quad (1)$$

The length of a smooth curve $\gamma : [0, 1] \rightarrow \mathcal{Z}$ connecting two points $\mathbf{z}_0 = \gamma(0)$ and $\mathbf{z}_1 = \gamma(1)$, so-called *geodesic*, is:

$$L(\gamma) = \int_0^1 \sqrt{\dot{\gamma}(t)^\top g(\gamma(t)) \dot{\gamma}(t)} dt, \quad (2)$$

where $\dot{\gamma}(t) = d\gamma/dt$ is the tangent vector along the curve. Unlike Euclidean distance, this length depends on the geometry of the space at every point along the path, naturally encoding regions of high and low transition resistance.

2.2 Pullback Metric

Rather than specifying $g(\mathbf{z})$ directly, we derive it by pulling back a metric from the reconstructed data space through the decoder $f_\theta : \mathcal{Z} \rightarrow \mathbb{R}^D$. Let $J_f(\mathbf{z}) \in \mathbb{R}^{D \times d}$ denote the Jacobian of the decoder at $\mathbf{z}$:

$$[J_f(\mathbf{z})]_{ij} = \frac{\partial [f_\theta(\mathbf{z})]_i}{\partial z_j}. \quad (3)$$

The standard pullback metric is:

$$g(\mathbf{z}) = J_f(\mathbf{z})^\top J_f(\mathbf{z}), \quad (4)$$

which measures distances in latent space by how much they stretch in data space under the decoder.

2.3 Refractive Pullback Metric

The standard pullback metric treats the output space as isotropic—all directions of change are equally weighted. To encode the heterogeneous resistance of tumor subtype

landscape, we propose the *refractive pullback metric*:

$$g_{\text{ref}}(\mathbf{z}) = \frac{1}{S} \sum_{s=1}^S J_f^{(s)}(\mathbf{z})^\top M(\mathbf{z}) J_f^{(s)}(\mathbf{z}), \quad (5)$$

where $M(\mathbf{z}) \in \mathbb{R}^{D \times D}$ is a position-dependent output-space modulation matrix:

$$M(\mathbf{z}) = [\alpha \cdot n(\mathbf{z}) + (1 - \alpha)] I_D, \quad (6)$$

with $\alpha \in [0, 1]$ controlling the strength of the refractive modulation and $n(\mathbf{z})$ a scalar refractive index field encoding boundary proximity:

$$\phi(\mathbf{z}) = -(d_{(2)}(\mathbf{z}) - d_{(1)}(\mathbf{z})), \quad n(\mathbf{z}) = 2 (1 + \tanh(\beta \phi(\mathbf{z}))), \quad (7)$$

where $d_{(1)}(\mathbf{z})$ and $d_{(2)}(\mathbf{z})$ are the distances from $\mathbf{z}$ to the nearest and second-nearest subtype centroids respectively, $\phi(\mathbf{z})$ is large and positive near subtype boundaries where two centroids are equidistant, and $\beta > 0$ controls the sharpness of the boundary transition. This construction is analogous to Snell’s law of refraction: near subtype boundaries the refractive index $n(\mathbf{z})$ is high, the metric inflates, and geodesics slow and bend—just as light slows and refracts at the interface between media of differing optical densities. The angle of refraction encodes the joint epigenomic-transcriptomic divergence between two phenotypic states—a steep bending of the geodesic at a subtype boundary signals a large regulatory discontinuity, where chromatin accessibility and gene expression programmes must undergo substantial coordinated reorganisation for the transition to proceed.

2.4 Geodesic Computation

A geodesic γ^* between two subtype centroids $\mathbf{z}_0$ and $\mathbf{z}_1$ is the length-minimising curve under the metric $g(\mathbf{z})$:

$$\gamma^* = \arg \min_{\gamma} L(\gamma) \quad \text{subject to} \quad \gamma(0) = \mathbf{z}_0, \quad \gamma(1) = \mathbf{z}_1. \quad (8)$$

This is equivalent to solving the geodesic ordinary differential equation (ODE), obtained from the Euler–Lagrange equations:

$$\ddot{\gamma}^k = -\Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j, \quad (9)$$

where Γ_{ij}^k are the Christoffel symbols of the second kind:

$$\Gamma_{ij}^k = \frac{1}{2} g^{kl} (\partial_i g_{jl} + \partial_j g_{il} - \partial_l g_{ij}), \quad (10)$$

and g^{kl} denotes the (k, l) entry of $g(\mathbf{z})^{-1}$. Partial derivatives $\partial_l g_{ij}$ are estimated via central finite differences:

$$\partial_l g_{ij}(\mathbf{z}) \approx \frac{g_{ij}(\mathbf{z} + \varepsilon \mathbf{e}_l) - g_{ij}(\mathbf{z} - \varepsilon \mathbf{e}_l)}{2\varepsilon}, \quad (11)$$

where $\mathbf{e}_l$ is the l -th standard basis vector and $\varepsilon > 0$ is a small step size. The geodesic ODE is rewritten as a first-order system:

$$\frac{d}{dt} \begin{pmatrix} \gamma \\ \dot{\gamma} \end{pmatrix} = \begin{pmatrix} \dot{\gamma} \\ -\Gamma_{ij}^k \dot{\gamma}^i \dot{\gamma}^j \end{pmatrix}, \quad (12)$$

and integrated numerically using an adaptive Runge–Kutta solver (RK45) with initial condition $\gamma(0) = \mathbf{z}_0$ and initial velocity $\dot{\gamma}(0) = \mathbf{z}_1 - \mathbf{z}_0$.

The geodesic velocity—the integrated arc-length under the metric—serves as a quantitative measure of transition resistance between subtype states:

$$V(\mathbf{z}_0, \mathbf{z}_1) = \int_0^1 \sqrt{\dot{\gamma}^*(t)^\top g(\gamma^*(t)) \dot{\gamma}^*(t)} dt. \quad (13)$$

Subtype pairs with low geodesic velocity are predicted to undergo facile transitions under the joint epigenomic-transcriptomic landscape, while high geodesic velocity indicates a deeply entrenched regulatory barrier. Waypoints of maximum metric curvature along the geodesic—positions where $\|g(\gamma^*(t))\|$ is largest—correspond to chromatin remodelling bottlenecks and are identified as candidate regulatory drivers of subtype switching.

2.5 Uncertainty Module

To allow for an uncertainty-aware pullback metric, one can define the following.

$$g_{\text{MC}}(\mathbf{z}) = J_f(\mathbf{z})^\top J_f(\mathbf{z}), \quad (14)$$

We estimate $J_f(\mathbf{z})$ via Monte Carlo (MC) dropout: by keeping dropout active at test time and averaging over S stochastic forward passes:

$$g_{\text{MC}}(\mathbf{z}) = \frac{1}{S} \sum_{s=1}^S J_f^{(s)}(\mathbf{z})^\top J_f^{(s)}(\mathbf{z}), \quad (15)$$

where each $J_f^{(s)}$ corresponds to a different dropout mask, encoding decoder uncertainty into the metric.

3 Results

In this section we provide an example of tumor subtyping trajectory analysis. We choose breast cancer as it is one of the most heterogeneous cases of tumor subtyping and can benefit largely from the trajectory analysis.

3.1 Simulated Breast Tumor Trajectory Analysis

Breast cancer consists of at least four tumor subtypes: Luminal A, Luminal B, Basal, and HER2-enriched. Luminal A tumors are estrogen-driven and slow; Luminal B are similar but faster-growing; Basal tumors are aggressive triple-negatives; HER2-enriched tumors overexpress a growth receptor that can be pharmacologically targeted [18]. Understanding how cells transition between these states and how similar or different they are in their underlying biology is a central question in cancer genomics. This experiment asks: can we learn a meaningful geometric map of these subtypes from their hormone-gene expression profiles, and use that geometry to measure how far apart the subtypes really are?

We simulate data for ten hormone-related genes [19, 20]—*ESR1*, *PGR*, *ERBB2*, *AR*, *ESR2*, *GPER1*, *IGF1R*, *INSR*, *PRLR*, and *OXTR*—measured across hundreds of BRCA tumour samples. These genes were chosen because they encode the receptors and signalling molecules that define the major subtype boundaries. As the task is to analyze the trajectory between subtypes, we simulate the rate of tumor plasticity.

We then train a Variational Autoencoder (VAE) to compress this ten-dimensional expression space into a four-dimensional latent space, and then reconstruct it. We perform the task of pulling back the geometry into the latent space by using the classical pullback metric and the refractive pullback metric that we proposed. For visualizing the trajectories we map the learned latents into two dimensions using PCA.

As shown in Fig. 1, Panel (a) shows the training loss converging smoothly over 300 epochs. The reconstruction is accurate; the model has learned a compact and faithful representation of the data. Panel (b) shows the result projected onto its first two principal components. The four subtypes separate cleanly: Luminal A sits in the upper right, Luminal B in the lower center, HER2 at the lower left, and Basal at the top left. Panel (c) confirms the biological grounding. The hormone gene profile heatmap shows that Luminal A and Luminal B share high *ESR1* and *PGR* expression, while HER2 is distinguished by its *ERBB2* signal [21], and Basal subtypes show a strikingly different profile. The geometry in (b) reflects the biology in (c): similar expression profiles cluster together, distant subtypes are far apart.

Euclidean distance in latent space is a naive measure of biological distance. It treats all directions equally, ignoring the fact that some directions in latent space correspond to large changes in gene expression while others correspond to almost none. A richer measure would weight distances by how much the decoder output actually changes as you move through latent space. This is the pullback metric. At every point z in latent space, the decoder defines a Jacobian J —a matrix describing how a small step in z translates into a step in the gene expression space. The pullback metric $G = J^T J$ is the inner product that measures distances in latent space as if you were

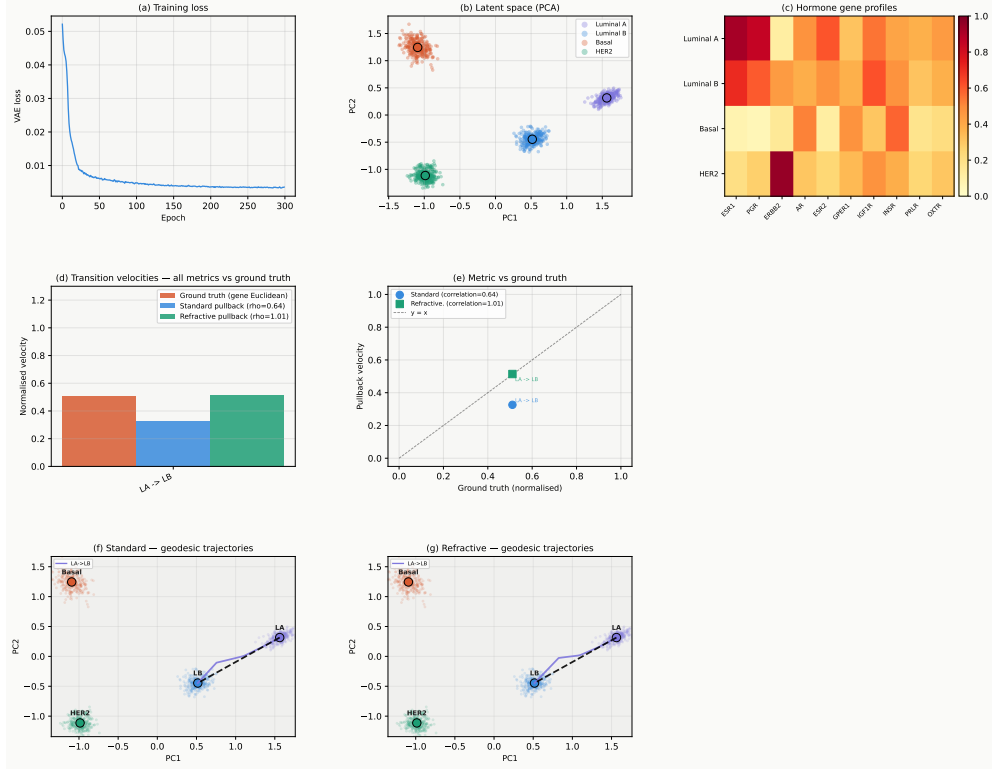


Fig. 1: Simulated data, their learned embedding by a VAE, and the trajectory analysis of the latent space for transition between Luminal A to Luminal B are shown.

actually measuring distances in gene expression space. To measure the cost of moving, we use the notion of velocity across transitions between two subtypes. In Fig. 1, we compute the transition velocity of Luminal A to Luminal B; in Panels d) and e), the comparison of the correlation to the ground truth is performed between pullback metric $G = J^T J$, referred to as standard, and our proposed refractive pullback metric $G = J^T M J$, where M enforces the subtype boundaries (the desired geometry) wrapped into the standard pullback metric. In both panels, our refractive metric (shown in green) exhibits a substantially higher correlation and closer agreement with the ground truth compared to the standard metric, and is in fact nearly the same as the ground truth.

Finally, in Fig. 1, Panels f) and g) show the trajectories from Luminal A (LA) to Luminal B (LB), using standard and refractive metrics, respectively. In each figure, the dashed line shows the Euclidean trajectory. It can be observed that Panel (g) exhibits a more pronounced curved geodesic trajectory than Panel (f), indicating a higher resistance to the LA-to-LB transition under the refractive metric compared to the standard metric. Moreover, both metric-induced trajectories are more realistic than the dashed Euclidean path, which assumes no resistance and thus allows transitions

between subtypes as easily as movements within a subtype. That is, the Euclidean path implies uniform energy across the space, failing to capture the increased cost associated with inter-subtype transitions.

4 Concluding Remarks

In this work, we propose using a pullback metric in the latent space of tumor subtypes to provide more plausible transitions between subtypes. Moreover, we propose a novel refractive metric that enforces resistance in the latent space when transitioning from one subtype to another. With breast cancer simulated data, we demonstrate that the geometry of a learned latent space, when properly equipped with a biologically informed metric, can recover meaningful measures of distance between tumor subtypes. The standard pullback metric captures some of the biological distance structure but systematically underestimates transition velocities. Our proposed refractive pullback metric achieves near-perfect agreement with ground-truth biological distances by weighting paths through regions of high hormonal activity more heavily.

As a future direction, the framework should be extended to investigate additional subtype transitions and evaluated across a broader range of tumor datasets.

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